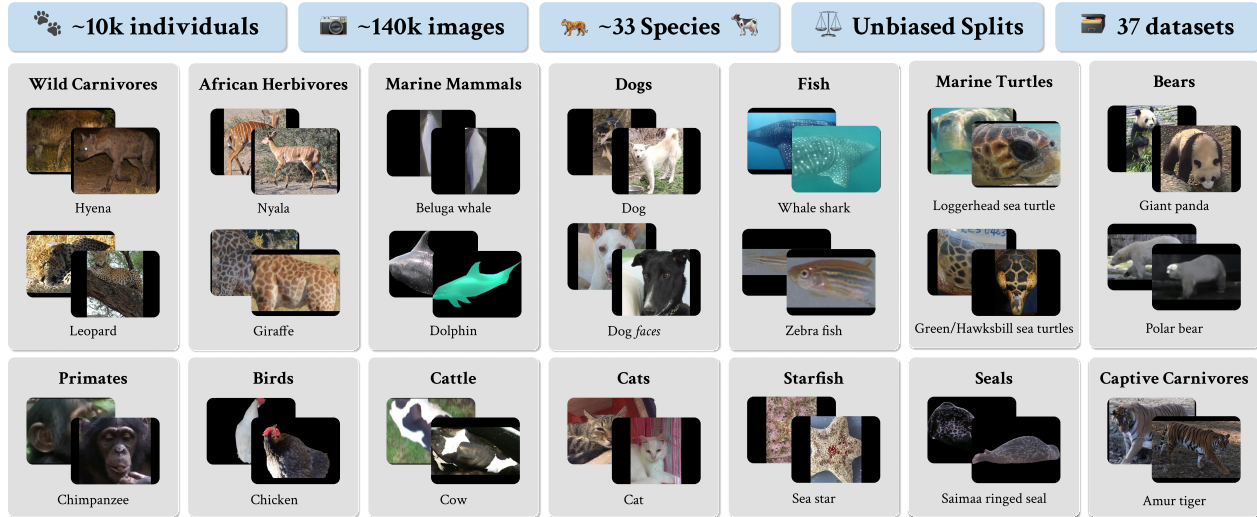


WildlifeReID-10k: Wildlife re-identification dataset with 10k individual animals

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Abstract

This paper introduces WildlifeReID-10k, a new large-scale re-identification benchmark with more than 10k animal identities of around 33 species across more than 140k images, re-sampled from 37 existing datasets. WildlifeReID-10k covers diverse animal species and poses significant challenges for SoTA methods, ensuring fair and robust evaluation through its time-aware and similarity-aware split protocol. The latter is designed to address the common issue of training-to-test data leakage caused by visually similar images appearing in both training and test sets. The WildlifeReID-10k dataset and benchmark are publicly available on Kaggle, along with strong baselines for both closed-set and open-set evaluation, enabling fair, transparent, and standardized evaluation of not just multi-species animal re-identification models.

1. Introduction

Individual animal re-identification assigns identities to individual animals in images. It is typically based on visual analysis of morphological characteristics, such as marks, spots, and stripes, which are stable over time and unique

to each individual. When combined with metadata (e.g., capture time and date, and location), this information helps study various aspects of wild animal populations’ biology and ecology. It can support tasks such as disease monitoring and control [39], assessing an animal’s role in the ecosystem [45], tracking invasive species [12], and evaluating human impact on habitats and ecological restoration efforts [10].

In recent years, there has been a significant increase in research on automatic re-identification methods for wild animals, resulting in the first-ever foundational models such as MegaDescriptor [54], MiewID [38], and WildFusion [13]. This surge has been made possible by recent advances in computer vision and machine learning and has also been driven by the growing availability of wildlife imagery obtained through various sources such as camera traps [46], drones [30], satellites [58], regular photo-identification surveys [48], and passive crowdsourcing from social media data [40]. This increase in research is evidenced not only by the increasing number of research studies [9, 11, 44] and the corresponding review papers [43, 46, 47, 55] but also by the expanding number of publicly available datasets for animal re-identification that cover numerous animal groups, including primates [19, 60], carnivores [16, 31, 52], reptiles [3, 17], whales [14], or mammals [2, 51, 63].

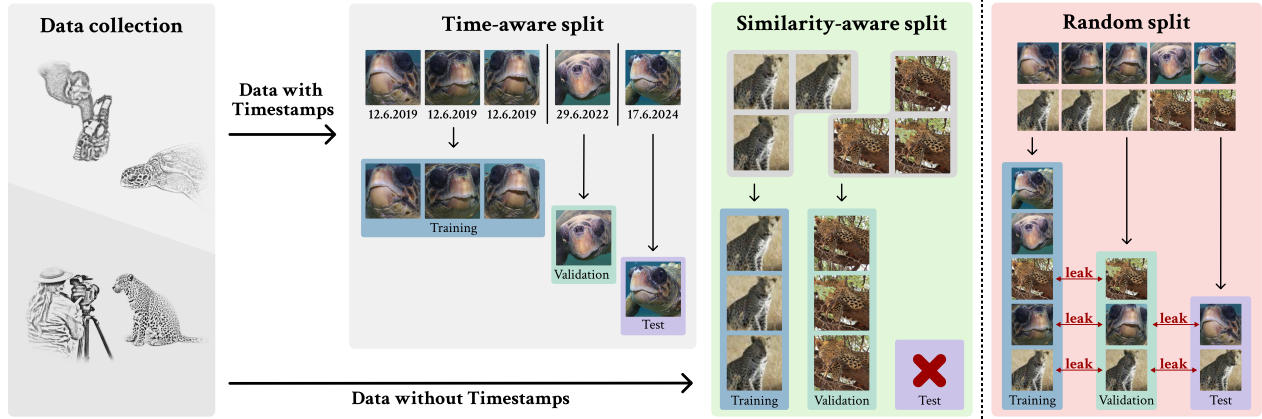


Figure 1. **WildlifeReID-10k splitting methodology.** Data collected in a single encounter is usually split randomly and might be *corrupted* by unwanted training-to-test set leakage. Therefore, we employ (i) a **time-aware split** if timestamps are available, and (ii) a **similarity-aware split**, where visually similar images are treated as a single observation which is assigned to either the training, validation, or test set.

However, many datasets have several limitations. Due to the high cost of labeling, most contain only a limited number of photos and individuals, reducing their standalone value. Documentation is often lacking, and formats are rarely unified, making initial analysis time-consuming. Even more critically, standardised training-testing splits and evaluation metrics are typically absent. As a result, numerous methods were often evaluated on only a subset of available datasets, with limited comparison to prior work [26, 47, 50].

Some of these issues were addressed by Cermak et al. [54], who released the `wildlife-datasets` library, which greatly simplifies dataset accessibility and handling. However, the library did not consider the "natural" open-set scenario where new identities are introduced, and does not address the training-to-test dataset leakage issues raised by Adam et al. [5]. Specifically, it is argued that the standard evaluation protocol, which randomly splits the dataset into training and testing sets, results in information leakage between the two. This leakage inflates performance metrics, as animal re-identification datasets often contain multiple similar photos taken during a single human-animal encounter¹ or extracted from consecutive video frames.

Therefore, while constructing WildlifeReID-10k, we employ a **time-aware split** [5] whenever timestamps are available, to prevent this leakage. More specifically, we select cutoff dates for each identity and assign all images captured before (after) this date to the training (validation or test) set. This guarantees that photos from the same encounter are placed either in the training or test set, but not both. Since most datasets do not have images with timestamps, we propose a new approach for encounter estimation by **visual similarity clustering**, where each cluster is treated as an observation, and all of its images are assigned to either the training or test set. See Fig. 1 for a visual explanation.

¹Usually one observation of an individual from one location and time.

WildlifeReID-10k is a curated animal re-identification benchmark that enables fair evaluation of models. Its relation to other existing dataset collections, such as PetFace [49], Wildlife-71 [26], or any private dataset collection, is complementary, since these collections can serve as additional data sources for training models. However, we recommend using WildlifeReID-10k for a fair comparison of model performance, which we hope will facilitate an unbiased measure of progress in animal re-identification.

The main contributions of this paper include:

- Splitting and evaluation methodology that prevents training-to-test data leakage, and allows fair and *less biased* evaluation without inflated performance.
- New dataset and benchmark – **WildlifeReID-10k** – for animal re-identification with more than 10k identities and 140k images of around 33 species.
- A set of baselines for both closed- and open-set settings.

2. Related work

Datasets: Publicly available datasets and tools have accelerated efforts to standardize and benchmark animal re-identification. A key contribution was made by Cermak et al. [54], who provided a thorough overview of existing datasets and introduced a unifying library (`wildlife-datasets`) that simplifies the dataset loading, preprocessing, and evaluation process, enabling fast, replicable, and comparable experiments across many species and scenarios. Following its success, several works introduced new datasets.

Jiao et al. introduced the Wildlife-71 dataset [26], using three existing datasets – ATRW [31], GiraffeZebraID [41], and SealID [36] – alongside manually labeled identities from the GOT-10k tracking dataset [25] and 816 YouTube videos. A significant part of Wildlife-71 is sourced from video datasets, which reduces the diversity of the images.

Shinoda et al. collected animal photos from pet shops and adoption websites and created the PetFace dataset [49]. Such an approach allows fast data collection of a large number of individuals, but raises potential ethical and privacy concerns. Moreover, over 90% of images are from indoors, i.e., not captured in the wild.

Methods: Most animal re-identification methods are developed and evaluated under a closed-set setting, where the identities present during testing are only those that have been included in the training set [13, 31, 38, 54]. This set-up simplifies the problem but fails to reflect real-world conditions, where animal populations are dynamic and new individuals are entering or leaving populations due to migration, birth, or other factors. As a result, models trained in a closed-set setting, despite their great performance, are limited in their ability to generalize to these unseen individuals.

Open-set recognition works under a more realistic setting as new identities are expected in the test set. Existing methods typically assign novelty scores based on softmax confidence, distance to decision boundaries, or embedding space proximity [23, 53, 56]. While widely used in other domains, this setting is still overlooked in animal re-identification. This gap highlights the need for more research into methods that generalize to novel individuals and support robust evaluation in open-set scenarios.

3. WildlifeReID-10k

The construction of the WildlifeReID-10k is built around the [wildlife-datasets](#) [54] framework – that unifies available wildlife re-identification datasets and provides an API that allows their straightforward access and download. WildlifeReID-10k includes 24 available datasets (bottom of Tab. 1) and extends the framework with 13 new datasets² (top of Tab. 1) that were not available earlier. In total, the dataset contains 140,488 images and 10,772 individuals of around 33 species. All images were either pre-cropped or cropped using the provided bounding boxes or segmentation masks, resulting in a dataset size of only 26 GB – or approximately 185 kB per image on average – without any resizing.

To further improve dataset quality, i.e., to prevent image repetition and to balance the ratio of images per individual, we selected every k -th frame from several datasets (*AAUZebraFish*, *AerialCattle2017*, *BirdIndividualID*, *MultiCamCows2024* $k = 20$; *PolarBearVidID* $k = 100$; *SMALST* $k = 10$). Without this, for instance *PolarBearVidID* would contain 140k images of just 13 individuals, and WildlifeReID-10k would be dominated by polar bears.

Additionally, we updated two datasets with new data (*BelugaID* and *SeaTurtleID2022*) and fixed several wrong labels in 3 additional datasets (*CTai*, *Cows2021*, *FriesianCattle2015*). In all datasets, we also found unidentified animals

(being named “Adult”, “new_whale” or “____”) which were also excluded. Consequently, around half of the data in WildlifeReID-10k (18/37 datasets) is either brand new or updated. Detailed scripts used to create WildlifeReID-10k are available through the [wildlife-datasets](#).

While there are several datasets with domestic animals, most of the animals are wild. Indeed, Tab. 1 implies that 50% of individuals were observed in the wild (71% when including rehabilitation centres). This number excludes wild animals in the zoos and confirms that the WildlifeReID-10k dataset indeed focuses mostly on wild animals.

Table 1. **Datasets used to create WildlifeReID-10k.** The (top) section lists newly added datasets, while the (bottom) section lists previously available datasets integrated into the collection.

	timestamp	images	identities	ratio
AmvrakikosTurtles [4]	✓	200	50	4.0
CatIndividualImages [32]	✗	13,021	509	25.6
Chicks4FreeID [27]	✗	1,146	50	22.9
CowDataset [20]	✓	1,485	13	114.2
DogFaceNet [35]	✗	8,363	1,393	6.0
NDD20 [51]	✗	2,657	82	32.4
MultiCamCows2024 [61]	✓	5,112	90	56.8
PolarBearVidID [62]	✗	1,391	13	107.0
PrimFace [1]	✗	1,282	68	18.9
ReunionTurtles [4]	✓	336	84	4.0
SeaStarReID2023 [57]	✓	2,187	95	23.0
SouthernProvinceTurtles [4]	✗	481	51	9.4
ZakynthosTurtles [4]	✓	160	40	4.0
AAUZebraFish [11]	✗	336	6	56.0
AerialCattle2017 [7]	✗	2,329	23	101.3
ATRW [31]	✗	5,415	182	29.8
BelugaID [2]	✓	8,559	788	10.9
BirdIndividualID [18]	✓	2,629	50	52.6
Cows2021 [21]	✓	8,670	179	48.4
CTai [19]	✗	4,662	71	65.7
CZoo [19]	✗	2,109	24	87.9
FriesianCattle2015 [6]	✗	193	25	7.7
FriesianCattle2017 [7]	✗	940	89	10.6
Giraffes [34]	✓	1,368	178	7.7
GiraffeZebraID [41]	✓	6,898	2,051	3.4
HyenaID2022 [52]	✗	3,129	256	12.2
IPanda50 [59]	✗	6,874	50	137.5
LeopardID2022 [52]	✗	6,806	430	15.8
MPDD [22]	✗	1,657	191	8.7
NyalaData [16]	✗	1,942	237	8.2
OpenCows2020 [8]	✗	4,736	46	103.0
SealID [36]	✗	2,080	57	36.5
SeaTurtleID2022 [5]	✓	8,729	438	19.9
SMALST [63]	✗	1,290	10	129.0
StripeSpotter [29]	✓	820	45	18.2
WhaleSharkID [24]	✗	7,693	543	14.2
ZindiTurtleRecall [3]	✗	12,803	2,265	5.7
TOTAL		13 140,488	10,772	13.0

²All selected datasets have permissible licenses for being re-uploaded.

3.1. Construction of default splits

The inconsistent training/test data splitting and evaluation methodology is a significant limitation to further animal re-identification research. A commonly used approach is the closed-set setting, where all individuals present in the test set have already been observed during training. In this scenario, each new image is of the known identities, and the assumption is that no other individuals will appear. This suits controlled environments, such as farms or zoos, where the population is fully known and static. However, this assumption rarely holds in wildlife monitoring. In contrast, the open-set setting acknowledges that new data may include previously unobserved individuals, such as newly born animals or individuals not yet included in the dataset.

Another methodological issue arises from the way that training and test sets are constructed. Developers often split images randomly without accounting for the structure of wildlife data. These datasets typically contain visually similar images, e.g., multiple photos taken during a human-animal encounter or consecutive frames extracted from videos. Random splitting can result in similar training and test images, leading to train-test leakage [5]. This leakage makes the classification task artificially simplified and can severely inflate reported performance, undermining the real-world utility of the proposed methods.

To the best of our knowledge, all available datasets either have no default split or use a random split without considering data acquisition details with only one exception (i.e., *SeaTurtleID2022*), which splits data based on timestamps – images taken before a specific date are assigned to the training set and those after are reserved for the test set. Hence, we propose a method for generating splits to mitigate training-to-test leakage. The method work as follows:

Step 1: Select open-set identities: To allow both open- and closed-set evaluation, we first select around 5% of all identities from each dataset, and we put all their images directly into the open-set test set.

Step 2: Time-aware clustering: From the remaining 95% we check whether the dataset contains timestamps, and if so, we split it in a time-aware fashion. Specifically, for each individual, we selected approximately 85% of the oldest observations and put them in the training set and the remaining into the test set. From datasets with timestamps (see Tab. 1), we excluded *AmvrakikosTurtles*, *ReunionTurtles*, *StripeSpotter*, and *ZakynthosTurtles* due to a limited number of images or encounters.

Step 3: Similarity-aware clustering: Most datasets lack timestamp metadata, making it difficult to determine encounters. Since images from the same encounter are often near-duplicates, they should lie close in the feature space of foundational models, such as DINOv2 [37] or CLIP [42].

Therefore, we first feed-forward all the data into the DINOv2 and store the resulting embeddings. To identify encounters, we use a variant of the single-linkage hierarchical clustering algorithm with a stopping criterion based on cosine similarity threshold θ ; in our case, set manually for each dataset to optimize the number of correct clusters (see Appendix 3.3 for more details). For each identity, we construct a graph where each image is a node, and an edge connects two nodes if their similarity is above the threshold. The connected components in this graph form our clusters. A more precise description can be found in Algorithm 1.

These clusters are then used to create the training-test split. For each identity, all its clusters are placed into the training set. The unclustered images are randomly divided into the training and test sets so that the training set contains the desired number of samples.

Algorithm 1 for finding clusters for the training-test split

Require: $S \in \mathbb{R}^{n \times n}$ similarity matrix, θ threshold

```

1: parent  $\leftarrow [0, 1, \dots, n - 1]$ 
2: Build and sort edges  $(i, j, S_{i,j})$  by descending similarity
3: function FIND( $i$ )
4:   while parent[ $i$ ]  $\neq i$  do
5:     parent[ $i$ ]  $\leftarrow$  parent[parent[ $i$ ]]
6:      $i \leftarrow$  parent[ $i$ ]
7:   end while
8:   return  $i$ 
9: end function
10: for  $(i, j, \text{sim})$  in sorted edges do
11:   if sim  $< \theta$  then break
12:   end if
13:   if Find( $i$ )  $\neq$  Find( $j$ ) then
14:     parent[Find( $i$ )]  $\leftarrow$  Find( $j$ )           ▷ merge clusters
15:   end if
16: end for
17: Build clusters by grouping nodes with same Find( $i$ ) root
18: return clusters

```

3.2. Dataset statistics

Following the above-mentioned approach, we formed two splits that are further described in Tab. 2. The closed-set setting contains 109,927 training images and 22,745 test images, with all test identities seen during training. In the open-set split, 523 new identities appear in the test set, resulting in 30,561 test images.

Table 2. **WildlifeReID-10k – proposed splits.** Values are grouped to highlight contrasts between closed- and open-set settings.

Setting	Images		Identities		
	Training	Test	Training	Test _{known}	Test _{new}
Closed-set	109,927	22,745	10,249	8,391	–
Open-set		30,561			523

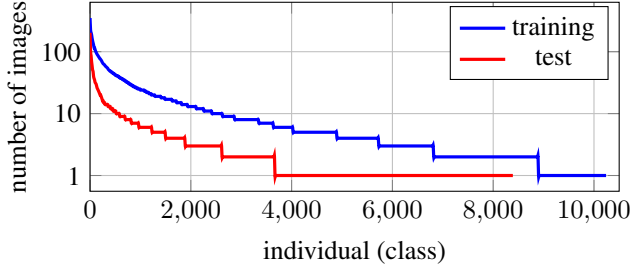


Figure 2. Number of images per individual in WildlifeReID-10k.

Approximately 56% of individuals in the test set have only one image, while the top 1% account for 19% of all images and the top 10% for 53%. This highlights a strong imbalance typical of datasets collected in the wild, where certain animals are captured more frequently – often because they frequent high-traffic areas like camera trap routes or shallow coastal waters. As a result, standard top-k accuracy metrics can be misleading, disproportionately favoring frequently observed and often well-studied individuals. To address this, we also report balanced accuracy, which accounts for such skew. In Fig. 2, we illustrate the distribution of images per individual in the training and test sets.

3.3. Quality of clusters

We can verify the clustering quality (i.e., purity) on datasets with timestamps. For each dataset, we design the *true-positives* TP and *false-positives* FP counts by:

$$TP = \sum_{c \in C} (|c| - 1), \quad FP = \sum_{c \in C} (|c| - n_{\max \text{ date}}(c)), \quad (1)$$

where C is the set of clusters. TP measures how many images were clustered above trivial singleton clusters. FP subtracts the number of images with the most frequent date from the cluster size. The clustering threshold θ controls a trade-off between cluster size and its quality.

On the one hand, the algorithm aims for large clusters per identity to address near duplicates in the dataset, which requires maximizing true positives (TP). On the other hand, overly large clusters may group images from different encounters, introducing noise. This shifts the task away from original identification by clustering not just duplicates but also images with true similarity features, which artificially increases the difficulty. Thus, false positives (FP) reflect cluster impurity and should be minimized.

In Figure 3 we show the dependence of TP and FP on the parameter θ . When θ is large, no clusters are formed, and both counts are 0. On the other hand, when θ is small, all images belong to one cluster, and both counts are large.

An optimal threshold keeps false positives (FP) low. In our case, with $\theta = 0.97$ we found 5,601 duplicates (TP) out of 45,637 images, with 685 incorrectly grouped (FP). However, since the datasets differ, using just one threshold

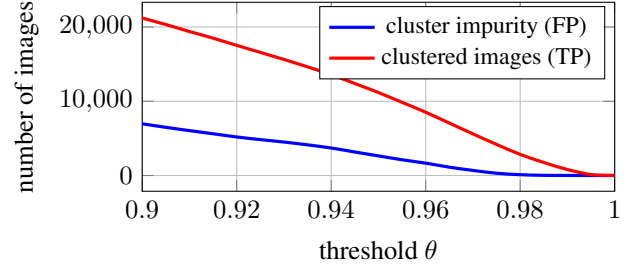


Figure 3. Size of clusters and their impurity based on θ .

is not ideal. To better account for dataset variability, one could manually select a threshold for each dataset based on visual inspection and improve the results significantly, i.e., got 8,206 TPs at 464 FPs in our case. Since the *manual* method was better, we used it for the similarity-aware split when timestamps were unavailable.

Table 3 and Figure 4 provide further evidence of the high quality of the obtained clusters. In the Table 3, rows indicate cluster size, columns the number of unique timestamps, and entries the number of clusters per size-timestamp combination. The final column shows the share of clusters with a single timestamp, indicating high-quality, time-consistent groupings. While small clusters almost always come from a single encounter, even the largest clusters maintain over 77% consistency.

Table 3. Number of clusters with certain sizes (rows) and unique timestamps (columns) for our clustering algorithm. The smaller the number of unique timestamps in a cluster, the higher quality of the corresponding cluster.

cluster size	unique timestamps						single time
	1	2	3	4	5	6+	
2	3009	78					97.5%
3	374	27	8				91.4%
4	193	26	4	2			85.8%
5	127	16	2	1	0		87.0%
6+	212	45	14	2	1	0	77.4%

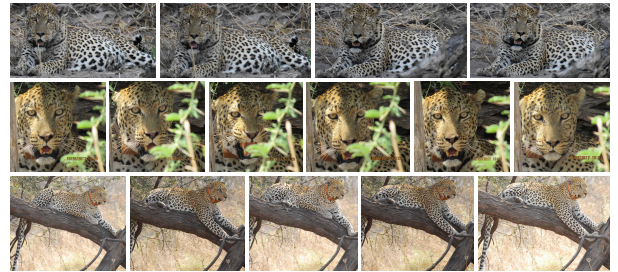


Figure 4. Found clusters (columns) in LeopardID2022 dataset. Most clusters consist of almost identical images with only small differences in size or the leopard’s head position.

4. Baselines

This section outlines the evaluation setup for both closed-set and open-set benchmarks. The closed-set setting measures model performance on known individuals and is suitable for controlled environments. The open-set setting reflects real-world dynamics, testing whether models can recognize unseen individuals and distinguish them from known ones.

4.1. Experimental Settings

To establish strong and robust baselines, we train models across both CNN and transformer-based architectures. Specifically, we use ConvNeXt-Base and EfficientNet-B3 as CNN models, and ViT-Base/p16 and Swin-Base/p4w7 as transformer-based alternatives.

While training, we follow the established methodology [38, 54] and optimize the models using metric learning with ArcFace loss [15] that aims to learn a feature extractor that produces highly discriminative feature embeddings. All models were trained for 100 epochs using SGD with momentum ($m=0.9$), cosine schedule ($10e^{-2}$ to $10e^{-4}$ and $10e^{-3}$ to $10e^{-6}$ for CNNs and Transformers respectively) and ArcFace with 64 scale and 0.5 margin.

Additionally, we train the same architectures but with cross-entropy loss. In this case, we used Adam optimizer [28] with cosine schedule from $10e^{-4}$ to $10e^{-6}$ for all.

Closed-set methodology. Models trained with ArcFace are used as feature extractors, and embeddings are computed for both training and test images. A 1-nearest neighbor classifier is constructed using the training embeddings, and test identities are predicted accordingly. For models trained with cross-entropy loss, predictions are made based on the index of the highest logit value.

Open-set methodology. ArcFace-trained models are evaluated using scores based on feature-space distance. Specifically, the Nearest Mean Score (NMS) [56], which measures the cosine distance to the centroid of the closest class. For cross-entropy models, the Maximum Softmax Probability (MSP) [23] and Maximum Logit Score (MLS) [53] are calculated, corresponding to the highest predicted probability and logit value, respectively.

Foundation models for animal re-id: To evaluate the generalization capabilities of existing foundation models for animal re-identification, we benchmark recent state-of-the-art feature extractors. Specifically, we use the pre-trained MiewID-msv3 model [38], which is designed for multi-species re-identification. We exclude MegaDescriptor-L [54] from our evaluation due to a substantial overlap between its training data and our test datasets, which would compromise the fairness of the comparison.

Despite a similar overlap with the training data of MiewID-msv3, we still include it in our experiments to assess its performance in real-world, non-curated scenarios and to evaluate the robustness of our proposed split strategies under such conditions. We evaluate the model in both **closed-** and **open-set** scenarios. In the closed-set setting, identification is done by retrieving the most similar training image based on cosine similarity. In the open-set setting, an image is classified as a new individual if its highest similarity score falls below a predefined threshold (see Eq. (3)).

To test the robustness of our proposed split strategy, we also include a **random split baseline**, ensuring the same number of training and test images per individual.

4.2. Metrics

Closed-set metrics: Closed-set classification is a well-defined problem, so standard metrics are used: top-1 (mTop1) and top-5 accuracy (mTop5), both macro-averaged across datasets to reflect expected image-level performance. Following Sec. 3.2, we also report balanced top-1 accuracy over datasets and individuals, referred to as BAKS (balanced accuracy on known samples) [33]. Mathematically:

$$\begin{aligned} \text{mTop1} &= \frac{1}{D} \sum_{d=1}^D \frac{1}{|I^d|} \sum_{i \in I^d} \mathbf{1}(\hat{y}_i = y_i), \\ \text{BAKS} &= \frac{1}{D} \sum_{d=1}^D \frac{1}{|C_d|} \sum_{c \in C_d} \frac{1}{|I^{d,c}|} \sum_{i \in I^{d,c}} \mathbf{1}(\hat{y}_i = y_i), \end{aligned} \quad (2)$$

where $\mathbf{1}$ is the indicator (0/1) function, D the number of datasets, C_d , the *known* classes (individuals) in dataset d and $I^{d,c}$ (or I^d) indices of all images of class c (or all classes) in the test set of the dataset d . Finally, y_i and $\hat{y}_i$ are the true and predicted identities of the i -th image.

Open-set classification metrics determine whether the individual is new (detection) or provide an identity if known (classification). The prediction is usually achieved by outputting a class c as well as a confidence score t that indicates how likely it is for the image to have a new identity. Whenever the confidence score is below some prescribed threshold t_{new} , the method predict the new class instead of c :

$$\text{prediction} = \begin{cases} c & \text{if } t \geq t_{\text{new}}, \\ \text{new individual} & \text{otherwise.} \end{cases} \quad (3)$$

The threshold t_{new} controls if a new individual is predicted, higher values equal a new individual, and lower values equal none, making open-set classification effectively a closed-set task with an additional class.

Open-set detection metrics: Detection of new individuals is a binary task, with known individuals treated as the

Table 4. **Closed-set performance on WildlifeReID-10k.** Top-1 and Top-5 accuracy and corresponding BAKS for different architectures and training methods. All metrics reported in %.

	Architecture	Input	Top1	Top5	BAKS
ArcFace	ConvNeXt-Base	224 ²	66.4	77.3	64.0
	EfficientNet-B3	300 ²	70.5	80.6	67.3
	ViT-Base/p16	224 ²	79.2	88.9	76.9
	Swin-Base/p4w7	224 ²	73.2	84.9	70.5
CCE	ConvNeXt-Base	224 ²	81.1	90.3	78.5
	EfficientNet-B3	300 ²	77.8	88.5	75.1
	ViT-Base/p16	224 ²	78.3	88.7	75.8
	Swin-Base/p4w7	224 ²	81.5	90.4	79.1

positive class. Detection metrics fall into two categories depending on whether they require specifying t_{new} . Some methods select this threshold automatically or average over all possible values. For instance, TNR@95TPR finds t_{new} such that at least 95% of known individuals are correctly identified (true positive rate, TPR), then reports the proportion of new individuals correctly detected (true negative rate, TNR). AUROC, on the other hand, computes the area under the ROC curve, which plots TPR against the false positive rate (FPR); since $\text{FPR} = 1 - \text{TNR}$, AUROC effectively aggregates $\text{TNR}@_{\alpha}\text{TPR}$ across all $\alpha \in [0, 1]$.

Balanced accuracy on unknown samples is an example of metrics requiring the threshold t_{new} , defined as

$$\text{BAUS} = \frac{1}{D} \sum_{d=1}^D \frac{1}{|C_d^*|} \sum_{c \in C_d^*} \frac{1}{|I^{d,c}|} \sum_{i \in I^{d,c}} \mathbf{1}(\hat{y}_i = \text{new}). \quad (4)$$

It is the same as BAKS, but the middle average is performed on the *unknown* classes C_d^* . As its name suggests, BAUS computes the balanced accuracy while considering only truly unknown individuals.

Combined metrics: To evaluate overall performance with a single score, we combine the closed-set (BAKS) and open-set (BAUS) metrics using their geometric mean:

$$\text{normalized accuracy} = \sqrt{\text{BAKS} \cdot \text{BAUS}}. \quad (5)$$

The geometric mean is preferred over the arithmetic mean, as it penalizes extreme imbalances, e.g., a trivial classifier predicting only new individuals yields $\text{BAKS} = 0\%$ and $\text{BAUS} = 100\%$, resulting in a misleading 50% arithmetic mean but a more appropriate geometric mean of 0%.

5. Results

This section presents an analysis of the performance of our closed-set and open-set baselines, followed by an evaluation of the MiewID foundational model.

Table 5. **Open-set performance on WildlifeReID-10k.** Various baseline architectures and scoring methods. AUROC and TNR@95TPR are reported in %.

Architecture	Method	Input	AUC	TNR@95
ConvNeXt-Base	MSP	224 ²	82.0	41.2
EfficientNet-B3		300 ²	81.2	36.3
ViT-Base/p16		224 ²	81.0	39.2
Swin-Base/p4w7		224 ²	82.9	44.4
ConvNeXt-Base	MLS	224 ²	82.2	41.9
EfficientNet-B3		300 ²	71.9	23.2
ViT-Base/p16		224 ²	81.5	39.7
Swin-Base/p4w7		224 ²	83.2	45.0
ConvNeXt-Base	NMS	224 ²	75.1	23.7
EfficientNet-B3		300 ²	76.7	32.6
ViT-Base/p16		224 ²	81.8	38.4
Swin-Base/p4w7		224 ²	76.7	33.0

Closed-set baselines: Table 4 compares the closed-set identification performance of various architectures trained with ArcFace and cross-entropy losses. Across all metrics, Top-1, Top-5, and BAKS, models trained with cross-entropy consistently outperform their ArcFace counterparts, except ViT, which also achieves the highest performance among ArcFace models with a Top-1 accuracy of 79.2% and a BAKS of 76.9%. However, the best overall results are obtained with cross-entropy trained Swin-Base, reaching 81.5% Top-1 accuracy, 90.4% Top-5, and a BAKS of 79.1%.

Open-set baselines: Consistent with the closed-set results, methods based on cross-entropy (MSP and MLS) generally outperform MSP, which is based on ArcFace. Among the cross-entropy approaches, MLS achieves the best results, with Swin reaching the highest AUROC (83.2%), and TNR@95 (45.0%). MSP also performs well with Swin (82.9% AUROC, 44.4% TNR). NMS lags behind across most architectures, except ViT. A more detailed description of open-set results is in Table 5.

MiewID performance: MiewID demonstrates strong performance across diverse datasets, but the evaluation split significantly impacts the results. As shown in Fig. 5 and detailed in Tab. 6, random splits can lead to overly optimistic outcomes, overestimating open-set performance by up to 12.3% on datasets with timestamps, due to data leakage.

In contrast, time- and similarity-aware splits introduce a higher difficulty, resulting in lower but more trustworthy performance metrics. These findings reinforce the importance of a split strategy for fair and meaningful evaluation.

We further analyze model behavior under varying thresholds t_{new} for identifying new individuals (see Figure 6). As expected, a high threshold favors BAUS while suppressing

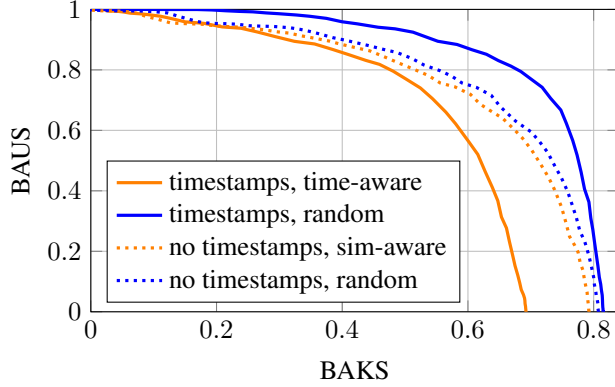


Figure 5. Performance difference between the proposed time-aware or similarity-aware split (orange) and the random split (blue) for datasets with timestamps (full lines) and without timestamps (dotted lines). Each point on a curve corresponds to a different threshold t_{new} . This figure suggests that the random split artificially inflates the performance of methods.

BAKS, and the reverse holds for low thresholds. The normalized accuracy, which balances the two, peaks at 65.1% and remains stable between 0.6 and 0.7; highlighting this range as a robust operating point for open-set identification.

Fig. 5 shows the performance of MiewID-msv3: on 9 datasets with timestamps, solid lines compare the time-aware split (orange) with the random split (blue); on 28 datasets without timestamps, dotted lines compare the similarity-aware split (orange) with the random split (blue).

Table 6. Performance on MiewID for the closed-set (BAKS) and open-set (area under BAKS-BAUS curve) settings.

Setting	timestamps		no timestamps	
	random	time-aware	random	sim-aware
<i>Closed-set</i>	81.5%	69.2%	80.8%	79.3%
<i>Open-set</i>	72.6%	55.9%	66.1%	63.6%

6. Conclusions

We introduced WildlifeReID-10k, the largest public dataset for animal re-identification, containing over 140,000 images of 10,772 individuals across 33 species. Unlike previous collections, it brings together a wide range of curated and standardized data in a single benchmark, with open-source tools for dataset creation, analysis, and evaluation.

A key contribution of this work is the introduction of realistic evaluation splits. We showed that commonly used random splits cause significant train-to-test leakage, leading to inflated performance. To address this, we proposed time-aware and similarity-aware splitting that better reflects the challenges of real-world wildlife monitoring. These are

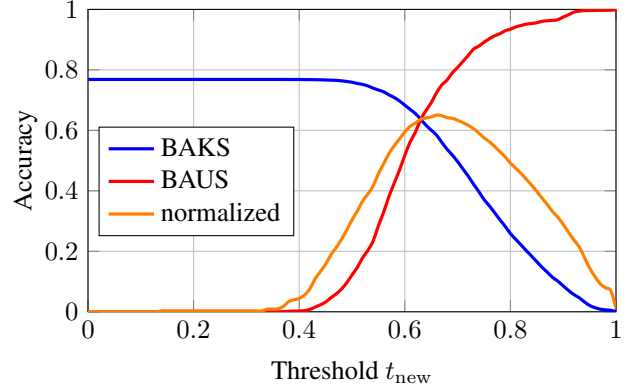


Figure 6. Performance of MiewID on the whole dataset for the open-set setting. The normalized accuracy is based on the threshold t_{new} for predicting a new individual. The geometric mean in (5) forces the normalized accuracy to be zero whenever BAKS or BAUS are zero.

integrated into the benchmark by default, enabling fair and reproducible evaluation.

Building on the [WildlifeDatasets](#) framework, we made several improvements: (i) 17 datasets were newly added or updated, (ii) low-quality datasets were removed and others significantly reduced in size, (iii) the total dataset size was reduced from 178GB to 26GB by removing redundant files and applying bounding boxes, (iv) multiple labeling errors were corrected, and (v) hosting the dataset on Kaggle removed the need for manual handling of individual datasets.

We believe WildlifeReID-10k will set a new standard in the field, offering a reliable foundation for training, testing, and comparing animal re-ID models across species. We hope it accelerates progress in both computer vision research and ecological applications, and invite the community to use, contribute to, and build upon this resource.

Limitations: While WildlifeReID-10k offers the most comprehensive and curated benchmark to date, there are still limitations. First, despite covering 33 species, many taxa remain underrepresented, especially amphibians and reptiles. Second, some datasets include only a few images per individual, which can make learning identity-specific features more difficult. Third, the similarity-aware split relies on visual clustering using pre-trained models, which may not always reflect true encounter boundaries. Finally, although we applied extensive cleaning, some label noise may persist due to the varied quality of the original datasets.

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References

- [1] The face images used in this study are provided by Prim-Face database: <http://visiome.neuroinf.jp/primface>, funded by Grant-in-Aid for Scientific research on Innovative Areas, "Face Perception and Recognition" from Ministry of Education, Culture, Sports, Science, and Technology (MEXT), Japan. 3
- [2] Beluga ID 2022, 2022. 1, 3
- [3] Turtle recall: Conservation challenge, 2022. 1, 3
- [4] Lukas Adam, Kostas Papafitsoros, Claire Jean, and Alan Rees. Exploiting facial side similarities to improve ai-driven sea turtle photo-identification systems. *bioRxiv*, pages 2024–09, 2024. 3
- [5] Lukáš Adam, Vojtěch Čermák, Kostas Papafitsoros, and Lukas Pícek. SeaTurtleID2022: A long-span dataset for reliable sea turtle re-identification. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 7146–7156, 2024. 2, 3, 4
- [6] William Andrew, Sion Hannuna, Neill Campbell, and Tilo Burghardt. Automatic individual holstein friesian cattle identification via selective local coat pattern matching in RGB-D imagery. In *2016 IEEE International Conference on Image Processing (ICIP)*, pages 484–488. IEEE, 2016. 3
- [7] William Andrew, Colin Greatwood, and Tilo Burghardt. Visual localisation and individual identification of holstein friesian cattle via deep learning. In *Proceedings of the IEEE International Conference on Computer Vision Workshops*, pages 2850–2859, 2017. 3
- [8] William Andrew, Jing Gao, Siobhan Mullan, Neill Campbell, Andrew W Dowsey, and Tilo Burghardt. Visual identification of individual holstein-friesian cattle via deep metric learning. *Computers and Electronics in Agriculture*, 185:106133, 2021. 3
- [9] Anka Bedetti, Cathy Greyling, Barry Paul, Jennifer Blondeau, Amy Clark, Hannah Malin, Jackie Horne, Ronny Makukule, Jessica Wilmot, Tammy Eggeling, et al. System for elephant ear-pattern knowledge (SEEK) to identify individual african elephants. *Pachyderm*, 61:63–77, 2020. 1
- [10] J David Blount, Mark W Chynoweth, Austin M Green, and Çağan H Şekercioğlu. COVID-19 highlights the importance of camera traps for wildlife conservation research and management. *Biological Conservation*, 256:108984, 2021. 1
- [11] Joakim Bruslund Haurum, Anastasija Karpova, Malte Pedersen, Stefan Hein Bengtson, and Thomas B Moeslund. Re-identification of zebrafish using metric learning. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision Workshops*, pages 1–11, 2020. 1, 3
- [12] Anthony Caravaggi, Marco Zaccaroni, Francesco Riga, Stéphanie C Schai-Braun, Jaimie TA Dick, W Ian Montgomery, and Neil Reid. An invasive-native mammalian species replacement process captured by camera trap survey random encounter models. *Remote Sensing in Ecology and Conservation*, 2(1):45–58, 2016. 1
- [13] Vojtěch Cermak, Lukas Pícek, Lukáš Adam, Lukáš Neumann, and Jiří Matas. Wildfusion: Individual animal identification with calibrated similarity fusion. *arXiv preprint arXiv:2408.12934*, 2024. 1, 3
- [14] Ted Cheeseman, Ken Southerland, Jinmo Park, Marilia Olio, Kiirsten Flynn, John Calambokidis, Lindsey Jones, Claire Garrigue, Astrid Frisch Jordan, Addison Howard, et al. Advanced image recognition: A fully automated, high-accuracy photo-identification matching system for humpback whales. *Mammalian Biology*, 102(3):915–929, 2022. 1
- [15] Jiankang Deng, Jia Guo, Niannan Xue, and Stefanos Zafeiriou. Arcface: Additive angular margin loss for deep face recognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 4690–4699, 2019. 6
- [16] Nkosikhona Dlamini and Terence L van Zyl. Automated identification of individuals in wildlife population using siamese neural networks. In *2020 7th International Conference on Soft Computing & Machine Intelligence (ISCMI)*, pages 224–228. IEEE, 2020. 1, 3
- [17] Stephen G Dunbar, Edward C Anger, Jason R Parham, Colin Kingen, Marsha K Wright, Christian T Hayes, Shahnaj Safi, Jason Holmberg, Lidia Salinas, and Dustin S Baumbach. Hotspotter: Using a computer-driven photo-id application to identify sea turtles. *Journal of Experimental Marine Biology and Ecology*, 535:151490, 2021. 1
- [18] André C Ferreira, Liliana R Silva, Francesco Renna, Hanja B Brandl, Julien P Renoult, Damien R Farine, Rita Covas, and Claire Doutrelant. Deep learning-based methods for individual recognition in small birds. *Methods in Ecology and Evolution*, 11(9):1072–1085, 2020. 3
- [19] Alexander Freytag, Erik Rodner, Marcel Simon, Alexander Loos, Hjalmar S Köhl, and Joachim Denzler. Chimpanzee faces in the wild: Log-Euclidean CNNs for predicting identities and attributes of primates. In *German Conference on Pattern Recognition*, pages 51–63. Springer, 2016. 1, 3
- [20] Lili Fu and Gong He. Cow dataset, 2021. 3
- [21] Jing Gao, Tilo Burghardt, William Andrew, Andrew W Dowsey, and Neill W Campbell. Towards self-supervision for video identification of individual holstein-friesian cattle: The Cows2021 dataset. *arXiv preprint arXiv:2105.01938*, 2021. 3
- [22] Zhimin He, Jiangbo Qian, Diqun Yan, Chong Wang, and Yu Xin. Animal re-identification algorithm for posture diversity. In *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1–5. IEEE, 2023. 3
- [23] Dan Hendrycks and Kevin Gimpel. A baseline for detecting misclassified and out-of-distribution examples in neural networks. In *International Conference on Learning Representations*, 2017. 3, 6
- [24] Jason Holmberg, Bradley Norman, and Zaven Arzoumanian. Estimating population size, structure, and residency time for whale sharks *Rhincodon typus* through collaborative photo-identification. *Endangered Species Research*, 7(1):39–53, 2009. 3
- [25] Lianghua Huang, Xin Zhao, and Kaiqi Huang. Got-10k: A large high-diversity benchmark for generic object tracking in the wild. *IEEE transactions on pattern analysis and machine intelligence*, 43(5):1562–1577, 2019. 2
- [26] Bingliang Jiao, Lingqiao Liu, Liying Gao, Ruiqi Wu, Guosheng Lin, Peng Wang, and Yanning Zhang. Toward re-

- identifying any animal. *Advances in Neural Information Processing Systems*, 36, 2024. 2
- [27] Daria Kern, Tobias Schiele, Ulrich Klauck, and Winfred Ingabire. Towards automated chicken monitoring: Dataset and machine learning methods for visual, noninvasive reidentification. *Animals*, 15(1):1, 2024. 3
- [28] Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014. 6
- [29] Mayank Lahiri, Chayant Tantipathananandh, Rosemary Warungu, Daniel I Rubenstein, and Tanya Y Berger-Wolf. Biometric animal databases from field photographs: identification of individual zebra in the wild. In *Proceedings of the 1st ACM international conference on multimedia retrieval*, pages 1–8, 2011. 3
- [30] Sarah Soffa Landeo-Yauri, Eric Angel Ramos, Delma Nataly Castelblanco-Martínez, Carlos Alberto Niño-Torres, and Linda Searle. Using small drones to photo-identify antillean manatees: A novel method for monitoring an endangered marine mammal in the caribbean sea. *Endangered Species Research*, 41:79–90, 2020. 1
- [31] Shuyuan Li, Jianguo Li, Hanlin Tang, Rui Qian, and Weiyao Lin. ATRW: A benchmark for Amur tiger re-identification in the wild. In *Proceedings of the 28th ACM International Conference on Multimedia*, page 2590–2598. Association for Computing Machinery, 2020. 1, 2, 3
- [32] Tzu-Yuan Lin. Cat individual images, 2020. 3
- [33] Pedro R Mendes Júnior, Roberto M De Souza, Rafael de O Werneck, Bernardo V Stein, Daniel V Pazinato, Waldir R de Almeida, Otávio AB Penatti, Ricardo da S Torres, and Anderson Rocha. Nearest neighbors distance ratio open-set classifier. *Machine Learning*, 106(3):359–386, 2017. 6
- [34] Vincent Miele, Gaspard Dussert, Bruno Spataro, Simon Chamailé-Jammes, Dominique Allainé, and Christophe Bonenfant. Revisiting animal photo-identification using deep metric learning and network analysis. *Methods in Ecology and Evolution*, 12(5):863–873, 2021. 3
- [35] Guillaume Mougéot, Dewei Li, and Shuai Jia. A deep learning approach for dog face verification and recognition. In *PRICAI 2019: Trends in Artificial Intelligence: 16th Pacific Rim International Conference on Artificial Intelligence, Cuvu, Yanuca Island, Fiji, August 26-30, 2019, Proceedings, Part III 16*, pages 418–430. Springer, 2019. 3
- [36] Ekaterina Nepovinnikh, Tuomas Eerola, Vincent Biard, Piia Mutka, Marja Niemi, Mervi Kunasranta, and Heikki Kälviäinen. SealID: Saimaa ringed seal re-identification dataset. *Sensors*, 22(19):7602, 2022. 2, 3
- [37] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. DINOv2: Learning robust visual features without supervision. *arXiv preprint arXiv:2304.07193*, 2023. 4
- [38] Lasha Otarashvili, Tamilselvan Subramanian, Jason Holmberg, JJ Levenson, and Charles V Stewart. Multispecies animal re-id using a large community-curated dataset. *arXiv preprint arXiv:2412.05602*, 2024. 1, 3, 6
- [39] Pablo Palencia, Rachele Vada, Stefania Zanet, Mara Calvini, Andrea De Giovanni, Giacomo Gola, and Ezio Ferroglio. Not just pictures: Utility of camera trapping in the context of african swine fever and wild boar management. *Transboundary and Emerging Diseases*, 2023:1–9, 2023. 1
- [40] Kostas Papafitsoros, Lukáš Adam, and Gail Schofield. A social media-based framework for quantifying temporal changes to wildlife viewing intensity. *Ecological Modelling*, 476: 110223, 2023. 1
- [41] Jason Remington Parham, Jonathan Crall, Charles Stewart, Tanya Berger-Wolf, and Daniel Rubenstein. Animal population censusing at scale with citizen science and photographic identification. In *2017 AAAI Spring Symposium Series*, 2017. 2, 3
- [42] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*, pages 8748–8763. PMLR, 2021. 4
- [43] Prashanth C Ravoor and TSB Sudarshan. Deep learning methods for multi-species animal re-identification and tracking—a survey. *Computer Science Review*, 38:100289, 2020. 1
- [44] Vito Renò, Giovanni Dimauro, G Labate, Ettore Stella, Carmelo Fanizza, Giulia Cipriano, Roberto Carlucci, and Rosalia Maglietta. A SIFT-based software system for the photo-identification of the Risso’s dolphin. *Ecological informatics*, 50:95–101, 2019. 1
- [45] J Marcus Rowcliffe, Juliet Field, Samuel T Turvey, and Chris Carbone. Estimating animal density using camera traps without the need for individual recognition. *Journal of Applied Ecology*, pages 1228–1236, 2008. 1
- [46] Stefan Schneider, Graham W Taylor, Stefan Linquist, and Stefan C Kremer. Past, present and future approaches using computer vision for animal re-identification from camera trap data. *Methods in Ecology and Evolution*, 10(4):461–470, 2019. 1
- [47] Stefan Schneider, Graham W Taylor, and Stefan C Kremer. Similarity learning networks for animal individual re-identification: an ecological perspective. *Mammalian Biology*, pages 1–16, 2022. 1, 2
- [48] Gail Schofield, Marcel Klaassen, Kostas Papafitsoros, Martin K. S. Lilley, Kostas A. Katselidis, and Graeme C. Hays. Long-term photo-id and satellite tracking reveal sex-biased survival linked to movements in an endangered species. *Ecology*, 101(7):e03027, 2020. 1
- [49] Risa Shinoda and Kaede Shiohara. Petface: A large-scale dataset and benchmark for animal identification. In *European Conference on Computer Vision*, pages 19–36. Springer, 2024. 2, 3
- [50] Ankita Shukla, Gullal Singh Cheema, Pei Gao, Suguru Onda, Divyam Anshumaan, Saket Anand, Ryan Farrell, et al. A hybrid approach to tiger re-identification. In *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops*, 2019. 2
- [51] Cameron Trotter, Georgia Atkinson, Matt Sharpe, Kirsten Richardson, A Stephen McGough, Nick Wright, Ben Burville, and Per Berggren. NDD20: A large-scale few-shot dolphin dataset for coarse and fine-grained categorisation. *arXiv preprint arXiv:2005.13359*, 2020. 1, 3

- [52] Botswana Predator Conservation Trust. Panthera pardus csv custom export, 2022. [1](#), [3](#)
- [53] Sagar Vaze, Kai Han, Andrea Vedaldi, and Andrew Zisserman. Open-set recognition: A good closed-set classifier is all you need. In *International Conference on Learning Representations*, 2022. [3](#), [6](#)
- [54] Vojtěch Čermák, Lukáš Pícek, Lukáš Adam, and Kostas Papafitsoros. Wildlifedatasets: An open-source toolkit for animal re-identification. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 5953–5963, 2024. [1](#), [2](#), [3](#), [6](#)
- [55] Maxime Vidal, Nathan Wolf, Beth Rosenberg, Bradley P Harris, and Alexander Mathis. Perspectives on individual animal identification from biology and computer vision. *Integrative and comparative biology*, 61(3):900–916, 2021. [1](#)
- [56] Tomáš Vojří, Jan Šochman, and Jiří Matas. Pixood: Pixel-level out-of-distribution detection. In *European Conference on Computer Vision*, pages 93–109. Springer, 2024. [3](#), [6](#)
- [57] Oscar Wahltinez and Sarah J Wahltinez. An open-source general purpose machine learning framework for individual animal re-identification using few-shot learning. *Methods in Ecology and Evolution*, 15(2):373–387, 2024. [3](#)
- [58] Jake Wall, George Wittemyer, Brian Klinkenberg, and Iain Douglas-Hamilton. Novel opportunities for wildlife conservation and research with real-time monitoring. *Ecological Applications*, 24(4):593–601, 2014. [1](#)
- [59] Le Wang, Rizhi Ding, Yuanhao Zhai, Qilin Zhang, Wei Tang, Nanning Zheng, and Gang Hua. Giant panda identification. *IEEE Transactions on Image Processing*, 30:2837–2849, 2021. [3](#)
- [60] Claire L Witham. Automated face recognition of rhesus macaques. *Journal of Neuroscience Methods*, 300:157–165, 2018. [1](#)
- [61] Phoenix Yu, Tilo Burghardt, Andrew W Dowsey, and Neill W Campbell. Multicamcows2024—a multi-view image dataset for ai-driven holstein-friesian cattle re-identification on a working farm. *arXiv preprint arXiv:2410.12695*, 2024. [3](#)
- [62] Matthias Zuerl, Richard Dirauf, Franz Koeferl, Nils Steinlein, Jonas Sueskind, Dario Zanca, Ingrid Brehm, Lorenzo von Fersen, and Bjoern Eskofier. Polarbearvidid: A video-based re-identification benchmark dataset for polar bears. *Animals*, 13(5):801, 2023. [3](#)
- [63] Silvia Zuffi, Angjoo Kanazawa, Tanya Berger-Wolf, and Michael J Black. Three-D safari: Learning to estimate zebra pose, shape, and texture from images "In the wild". In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 5359–5368, 2019. [1](#), [3](#)